

8. (a) (i) The table below shows results of tests carried out on three white Group 1 compounds, **A**, **B** and **C**.

The flame tests were carried out on solids and the silver nitrate tests on solutions of the compounds.

Test	Compound A	Compound B	Compound C
Flame test	red	yellow	lilac
Add silver nitrate solution	white precipitate	cream precipitate	yellow precipitate

Use the information in the table to identify the compounds.

[2]

Compound **A**

Compound **B**

Compound **C**

- (ii) The symbol equation represents the reaction occurring between solutions of silver nitrate and magnesium chloride.

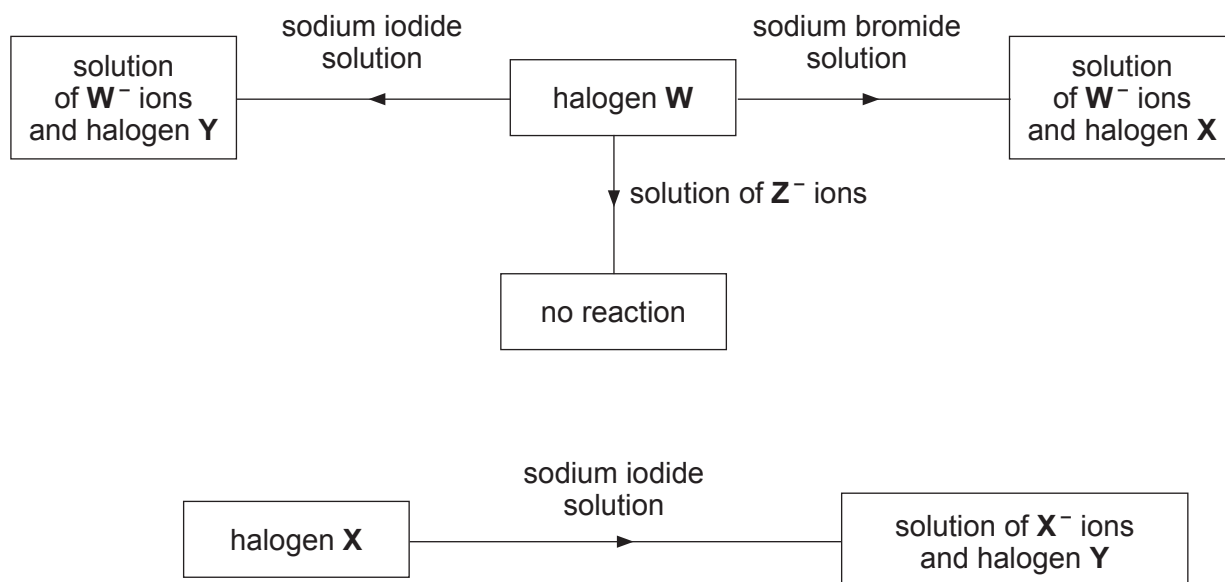


Write the **ionic** equation for the reaction. Include state symbols in your answer. [3]

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- (b) **W**, **X**, **Y** and **Z** represent the halogens fluorine, chlorine, bromine and iodine, but not necessarily in that order.

The diagrams below show some reactions of halogens **W** and **X**.



Use the information to identify halogens **W**, **X**, **Y** and **Z**.

[2]

W

X

Y

Z

- (c) Iron wool burns in fluorine to give iron(III) fluoride.

Write a balanced **symbol** equation for this reaction.

[2]

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